

Training an Agent-Trajectory Retriever

Final-Defense Proposal · v3.0

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Abstract

In Agentic Search, an agent issues intermediate queries, examines snippets, chooses which documents to browse in full, and uses the resulting evidence in later reasoning. LRAT identifies Search-to-Browse transitions, unbrowsed candidates in the same search context, and post-Browse reasoning as useful signals of document utility. Our project did not regenerate agent trajectories, re-run the paper’s judge, or introduce external data, teacher parameters, LoRA, or adapters. Instead, we built an auditable training and delivery path around the official Training pairs, Agent trajectories, offline Corpus, and the specified Qwen3-Embedding-0.6B. The final A-leaderboard result is Total 40.6, Recall 46.3, Success 21.1, and 16.0 average steps; the three-Agent B-leaderboard evaluation averages 56.18 Total.

Keywords: Agent retrieval; query-disjoint split; weighted contrastive learning; provenance audit; reproducible training

1 Motivation and Scope

Deep-research agents alternate among thinking, searching, browsing, and reasoning. Retrieval is therefore not only a relevance endpoint: it supplies evidence for the next reasoning step. Our implementation path is:

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fixed inputs -> trajectory/pair provenance audit -> normalized-query isolation
-> 1 positive + 5 negatives -> weighted training -> A/B evaluation
-> independent loading, freezing, and reproduction read-back
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The LRAT paper treats browsed documents as candidate positives, unbrowsed documents from the same Search candidate set as a negative pool, and post-Browse reasoning as a way to filter weak positives and express relevance intensity. We use the published Training pairs as the training supervision. Trajectories and Corpus records are used for provenance and document-consistency checks; they are not used here to collect new trajectories, recompute labels or weights, or claim completion of the paper’s Data Flywheel.

2 Provenance and Query Isolation

For every pair record, we trace its query, positive document, negative document references, and **reweight_rate** back to trajectory events and verify the referenced text against the fixed Corpus. Among 96,504 Training pairs, 96,366 are stable mappings, 138 are ambiguous but field-consistent mappings, and 0 are mismatches. The audit covers 1,989,015 document references and 959,042 unique document IDs, with no missing documents or text mismatches.

We strip leading and trailing whitespace, collapse repeated whitespace, and apply Unicode case folding. All source rows sharing a normalized query form one group and are assigned together:

Set	Query groups	Source rows	Role
Train	78,890	94,113	Full-parameter training
Dev	1,500	–	Development and selection
Locked test	500	–	Not used for training or selection

The three sets have zero normalized-query overlap. The locked test was not used for model, method, or hyperparameter selection.

3 Training Method

Each Training-pair row becomes one query, one positive passage, and five negatives randomly sampled from that row’s **neg** list; the positive is fixed at candidate position 0. The resulting six-way candidate group is an input construction unit, not the optimizer batch size. Query and passage caps are 128 and 512 tokens.

Starting from the specified Qwen3-Embedding-0.6B, we update all parameters with shared encoder weights, last-token pooling, L2 normalization, and 1024-dimensional representations. With temperature $\tau = 0.02$, query loss ℓ_j , and the published pair weight w_j , the implemented objective is

$$\mathcal{L} = \frac{\sum_j w_j \ell_j}{\sum_j w_j}.$$

The weight changes a query’s contribution to the loss, not its positive/negative label. Each of two A40 GPUs processes one candidate group. Same-order cross-device gathering makes the other query’s candidates in-batch negatives and keeps each weight attached to its query. Gradient accumulation is 4, giving an effective query batch of 8 without merging the candidate pools of four micro-steps.

Configuration	Value
Base model	Qwen3-Embedding-0.6B
Training rows	94,113
Hardware	$2 \times$ NVIDIA A40
Epochs / optimizer steps	1 / 11,764
Per-device batch / accumulation	1 / 4
Group size / learning rate	$6 / 2 \times 10^{-6}$
Query / passage cap	128 / 512
Precision / negatives	bf16 / cross-device

4 Configuration Selection

We first fixed the learning rate at 1×10^{-6} , trained on all 96,504 Training pairs for 1, 2, and 3 epochs, and submitted each model to the A leaderboard. Their Total scores were 40.1, 38.2, and 38.7, so we selected one epoch. We then fixed the 94,113-row query-disjoint train set and compared four learning rates for 500 steps on dev1500:

Epochs	Total	Recall	Success	Avg. steps
1	40.1	44.8	21.0	15.4
2	38.2	42.7	18.3	15.5
3	38.7	43.4	19.0	15.8

Learning rate	R@1	MRR
3×10^{-7}	0.4913	0.6419
5×10^{-7}	0.5073	0.6571
1×10^{-6}	0.5433	0.6831
2×10^{-6}	0.5640	0.6993

5 A/B Results and Boundaries

Candidate	Total $\uparrow$	Recall $\uparrow$	Success $\uparrow$	Avg. steps $\downarrow$
Qwen3-Embedding-0.6B (Base)	28.5	28.5	10.6	17.8
Official LRAT baseline	34.0	36.1	15.5	16.8
Query-disjoint (1×10^{-6})	39.5	45.0	19.8	15.9
Query-disjoint (2×10^{-6})	40.6	46.3	21.1	16.0

The final candidate is 12.1 Total points above the base and 6.6 above the official LRAT baseline. Within the same query-disjoint recipe, changing the learning rate moves Total from 39.5 to 40.6. These are comparisons between reported leaderboard rows and are not a single-variable causal attribution.

The B-leaderboard Totals are 67.42 for DeepSeek-V4-Flash-0731, 60.96 for Qwen3.5-35B-A3B, and 40.16 for gpt-oss-120b, averaging 56.18; the range is 27.26. This is direct cross-Agent runnability evidence. Because matched per-Agent baselines are not available, we report absolute results and do not claim universal improvement across agents.

6 Delivery and Conclusion

The final model passed fresh CPU loading, query/document inference smoke, freeze-manifest checks, anonymous Hugging Face revision read-back, and GitHub/reproduction-package validation. Detailed hashes, commits, and file-level evidence are kept in the reproduction appendix rather than the main narrative.

Our contribution is an auditable pipeline around fixed official supervision: provenance explains where the signals came from, query-disjoint grouping protects the development boundary, explicit candidate construction and weighted loss record the training semantics, staged experiments select the epoch and learning rate, and A/B evaluation reports both measured performance and its cross-Agent limits. The project does not present the competition training run as a complete re-execution of the LRAT paper’s trajectory collection, judge, or Data Flywheel.

References

1. Yuqi Zhou et al. “Learning to Retrieve from Agent Trajectories.” arXiv:2604.04949, 2026.
2. BC-Plus A Leaderboard.
3. CCIR Cup 2026 competition page.

A Delivery Evidence and Scope

The Hugging Face evidence is anonymous API read-back of the public commit, file table, file size, and LFS SHA. It is not described as a fresh local re-download and hash recomputation of the 2.38 GB weight.

Evidence layer	Recorded check	Result
Manifest	Base model, raw pairs, provenance, corpus, query-disjoint split, command, and model identity	Fixed and traceable through reproduce_m10.py , manifests, and the freeze record
Independent load	Fresh CPU process loads the frozen model; query/document inference smoke	Qwen3Model , 595,776,512 parameters, 1024-dimensional outputs
Canonical dry-run	Recheck provenance, corpus, train split, and the two-GPU command without training	dry_run_verified ; 96,504 provenance rows, 11,215,099 corpus rows, 94,113 train rows
HF public read-back	Anonymous API read-back of public commit, root file table, file size, and LFS SHA	Public repo; model.safetensors 2,383,139,480 bytes; LFS SHA equals the frozen checkpoint SHA
GitHub and ZIP	Repository checks plus package manifest, CRC, and validators	main@31949f6 ; tests, compile, JSON/diff checks, 45-file ZIP, CRC and package validators pass

The frozen identities are:

Artifact	Identity
Base model	Qwen3-Embedding-0.6B ; SHA-256 0437e45c94563b09e13cb7a64478fc406947a93cb34a7e05870fc8dcd48e23fd
Raw pairs	96,504 rows; SHA-256 dd75a3f1970438f0905a3e3e93e3d98dc1122cdb0e054ba87159a9368afbe1b9
Query-disjoint train	94,113 rows; SHA-256 158eb3843e8e022b5b0d7e64446ddf0782e2ad1aaa830f9f5aa3fb3b06c835c9
Provenance	96,366 stable / 138 ambiguous / 0 mismatch; SHA-256 0dd510fcc41444bfcff99381ce077e5048375855add8498f9ab10afabea4b690
Offline corpus	11,215,099 rows; SHA-256 4d795938bb89cbd7e7467a8da4e772f7ae95e6b533181aeace2a5e3fd3de6393
Final checkpoint	model.safetensors ; SHA-256 cea87cca521abd46c3e45ae53cd3b8f65b8216348c373fcc9698d69f133a9984
Freeze manifest	SHA-256 d56ceafcc110ea37d480d1f137b7306a63422afc1b4e257e44119e27ada11b2a
Hugging Face	Seraphic663/lrat-qwen3-0.6b-lr2e6-full-20260729 ; commit 9cdfdba51359ec65069a748cf8c00c55477016bb
GitHub	seraphic663/ccir-lrat-retriever ; main@31949f6f53722f06e91bd2ded6ec7f3a48037bba
B-leaderboard JSONL	versions/M10/submission_B_m10_20260731.jsonl ; SHA-256 02c271173ad6bf5e438972bf9e617c66e1625bb13f5a7086c1b6bcf0167b3f49
Reproduction ZIP	45 files, 2,387,249,735 bytes; SHA-256 c46f15d34a09dfe3108971ce56f837f1a23bbe39e35ee3a541bb609aea159984

These checks establish traceability and delivery-contract consistency. They do not guarantee a byte-identical checkpoint on every hardware and runtime combination, and API metadata read-back is not a local re-download and hash recomputation.